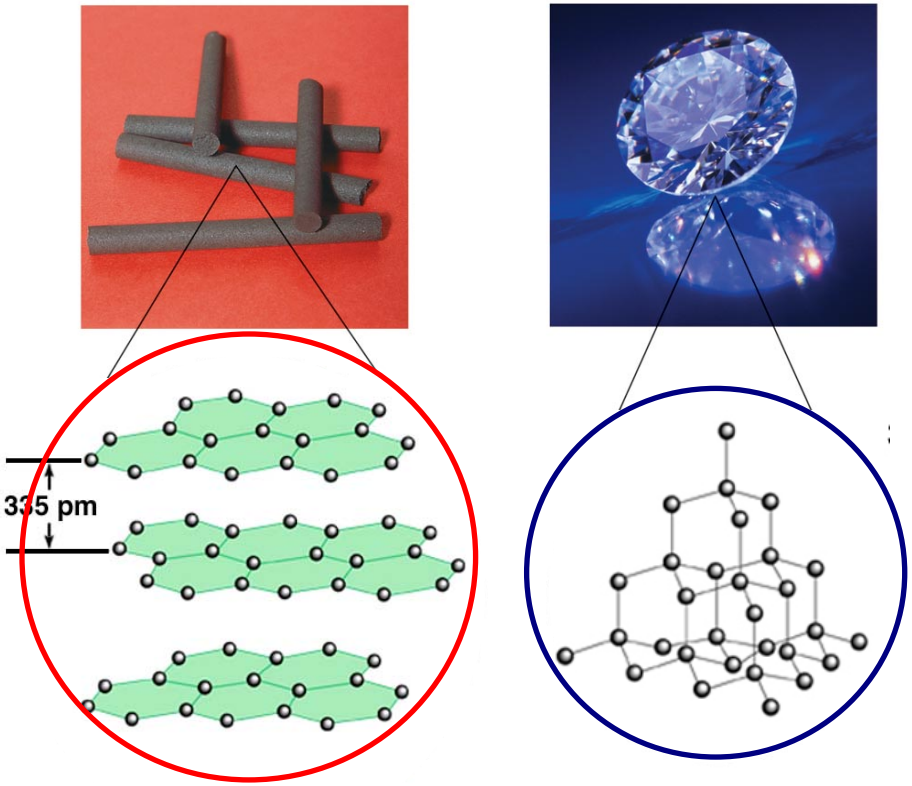
화학특강 (2009년 2학기)
제6장

황현석

2009년 11월 30일



Allotropes of Carbon – Graphite and Diamond



(left) graphite; (right) diamond.

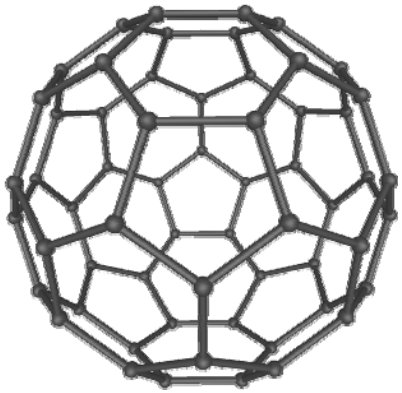
$C_{diamond}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°=-397$ kJ

$C_{graphite}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°=-394$ kJ

$C_{diamond}(s) \rightarrow C_{graphite}(s)$ $\Delta G° =?$ kJ

Kinetic control & Thermodynamic Control

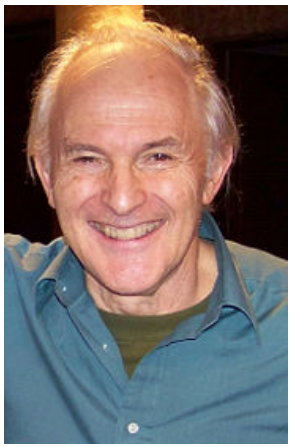
Allotropes of Carbon – Buckminsterfullerene (C_{60})



Buckminsterfullerene (6_{60})
or buckball



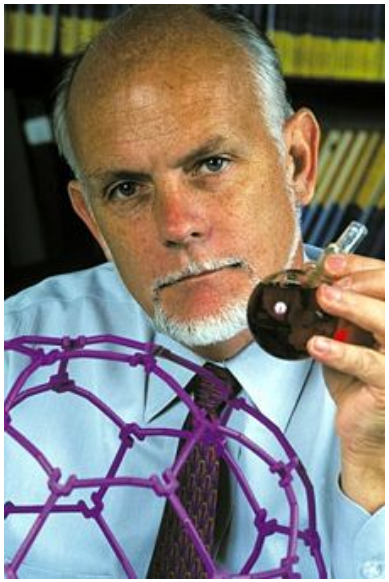
Geodesic dome by
R. Buckminster Fuller



Harold Kroto



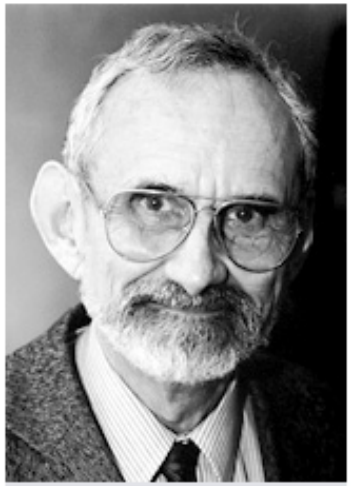
Robert Curl



Richard Smalley

The Nobel Prize in Chemistry 1996

"for their discovery of fullerenes"



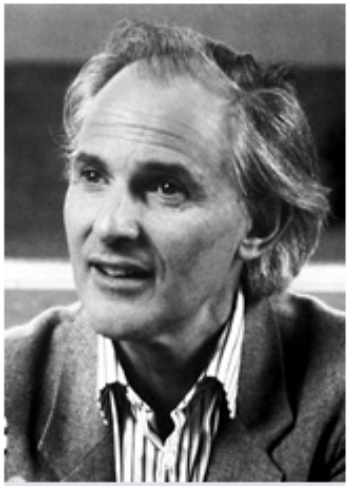
Robert F. Curl Jr.

1/3 of the prize

USA

Rice University
Houston, TX, USA

b. 1933



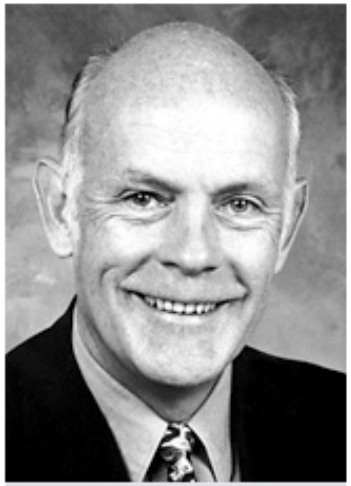
Sir Harold W. Kroto

1/3 of the prize

United Kingdom

University of Sussex
Brighton, United Kingdom

b. 1939



Richard E. Smalley

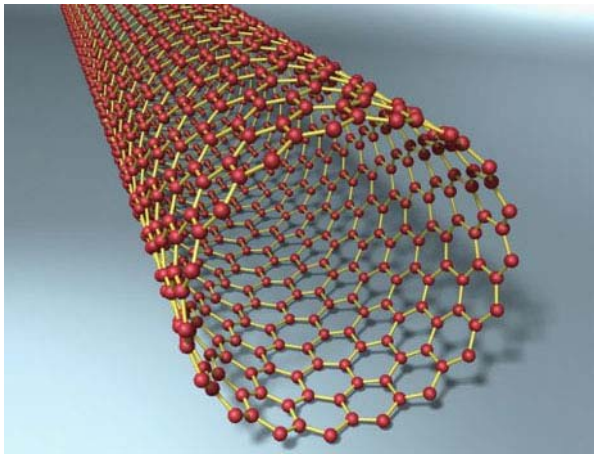
1/3 of the prize

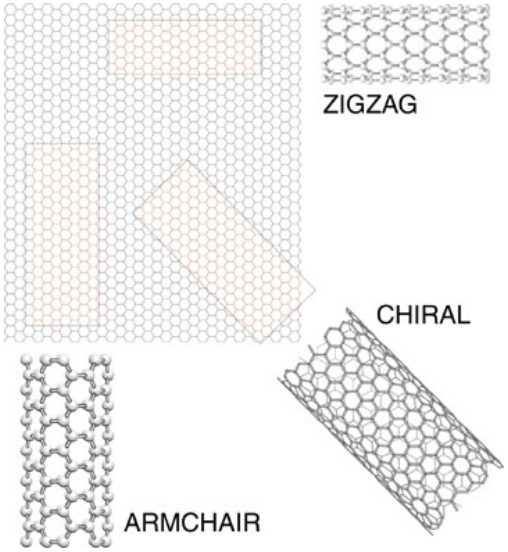
USA

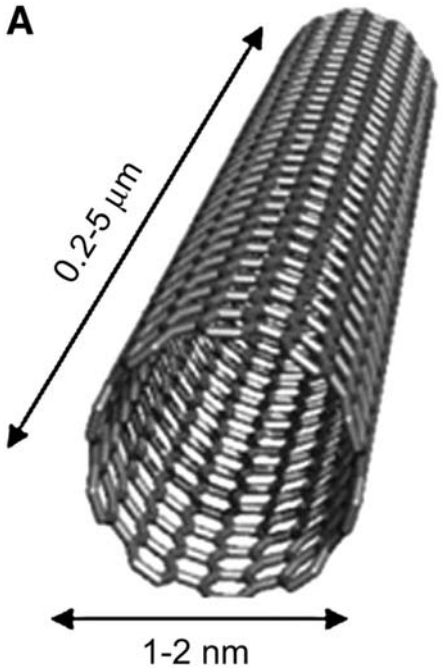
Rice University
Houston, TX, USA

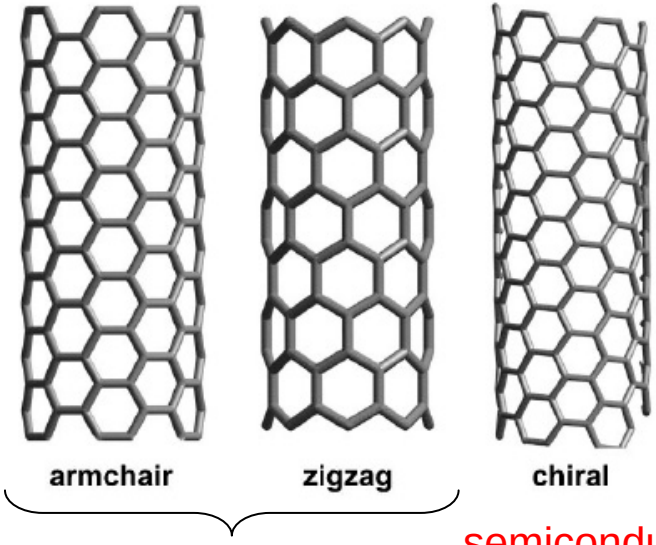
b. 1943
d. 2005

Allotropes of Carbon – Carbon Nanotube





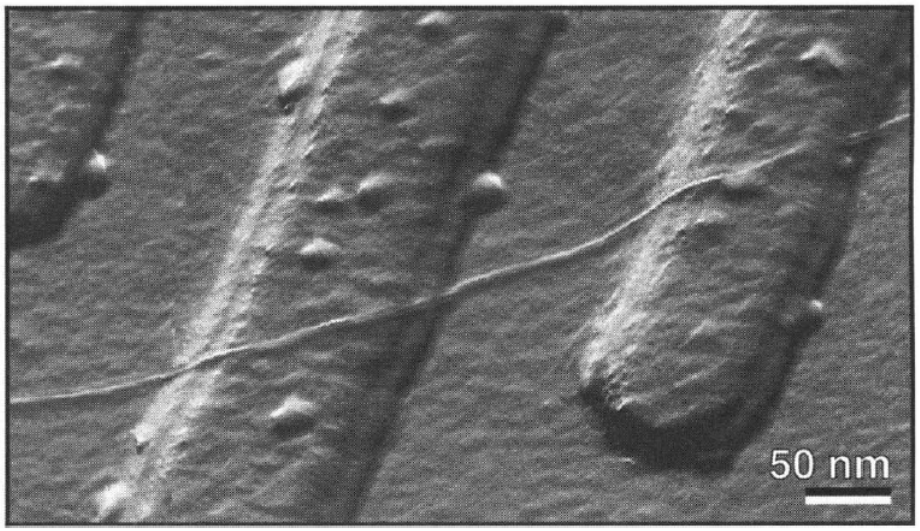




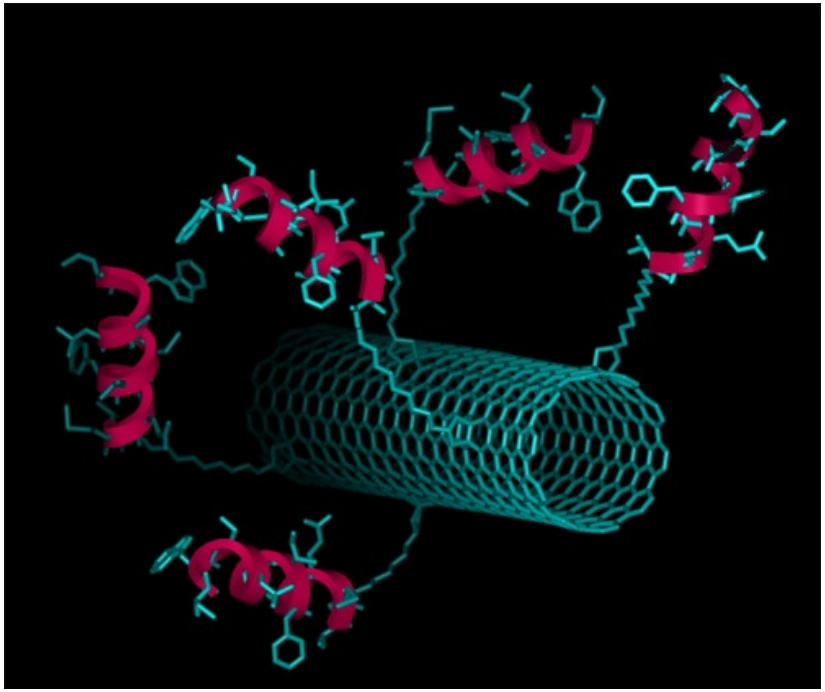
semiconductor

metallic, conductor

Applications of carbon nanotubes

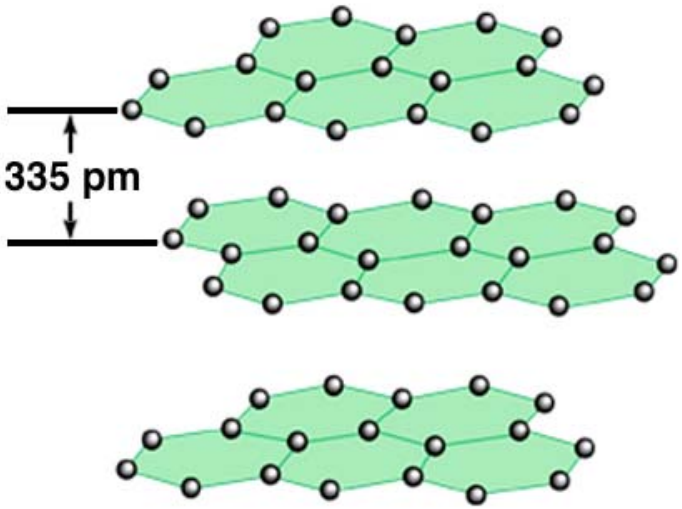


AFM tapping-mode image of a carbon nanotube
on top of a Si/SiO_2 substrate (Nature 386, 474 (1997)

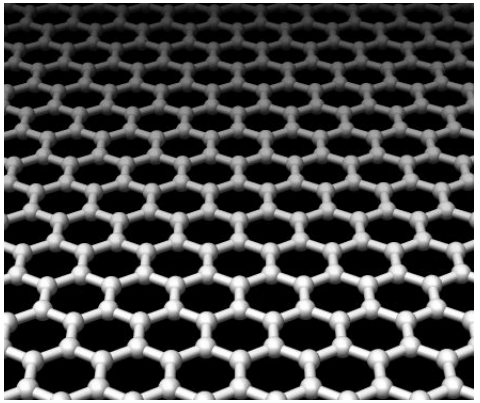


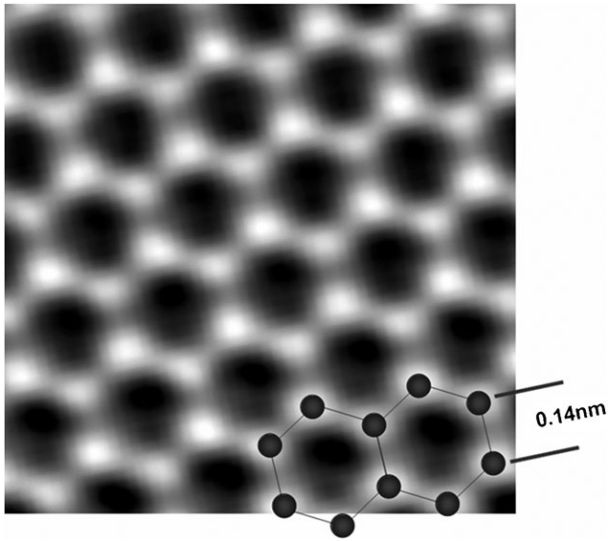
Allotropes of Carbon – Graphene

- Graphite



- Graphene (Science **306**, 666 (2004))





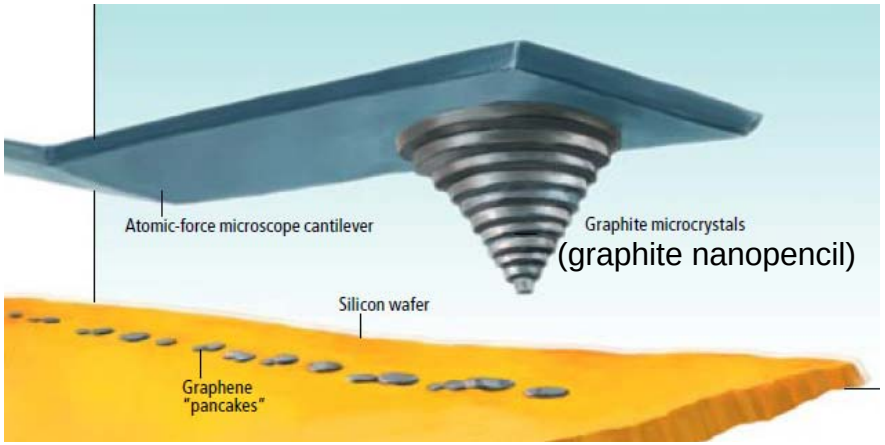
Transmission electron microscope (TEM) image of graphene

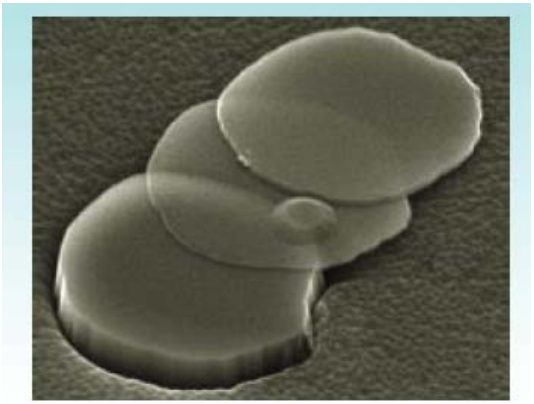
How to Make Graphenes

- Using a plastic sticky tape

Graphene Scratching 101

- Using a graphite nanopencil



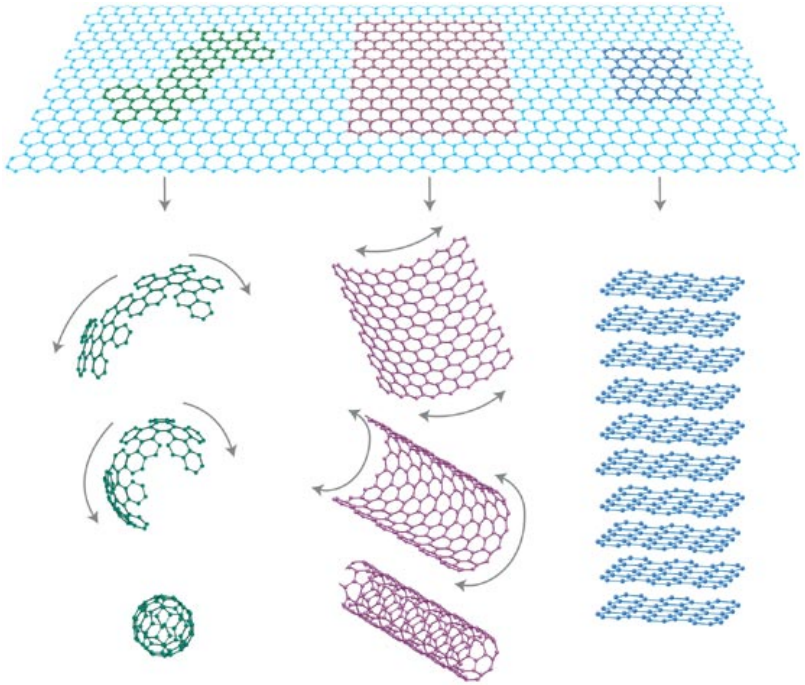


Thin graphene pancakes onto a silicon wafer

KNU

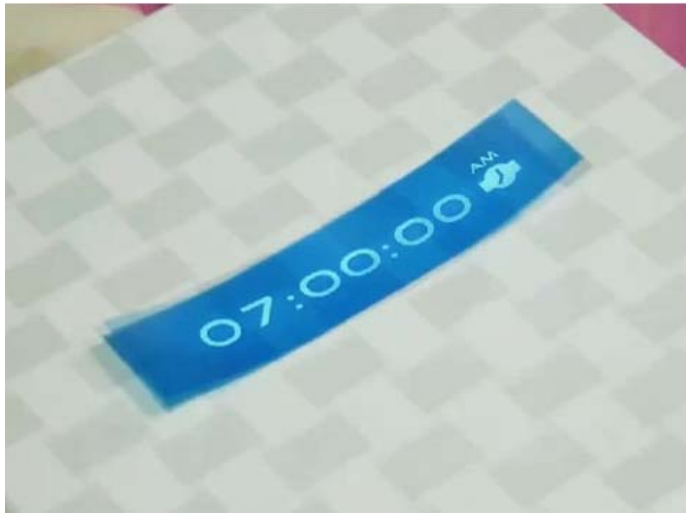
Applications of the Graphene

- Mother of most carbon allotropes

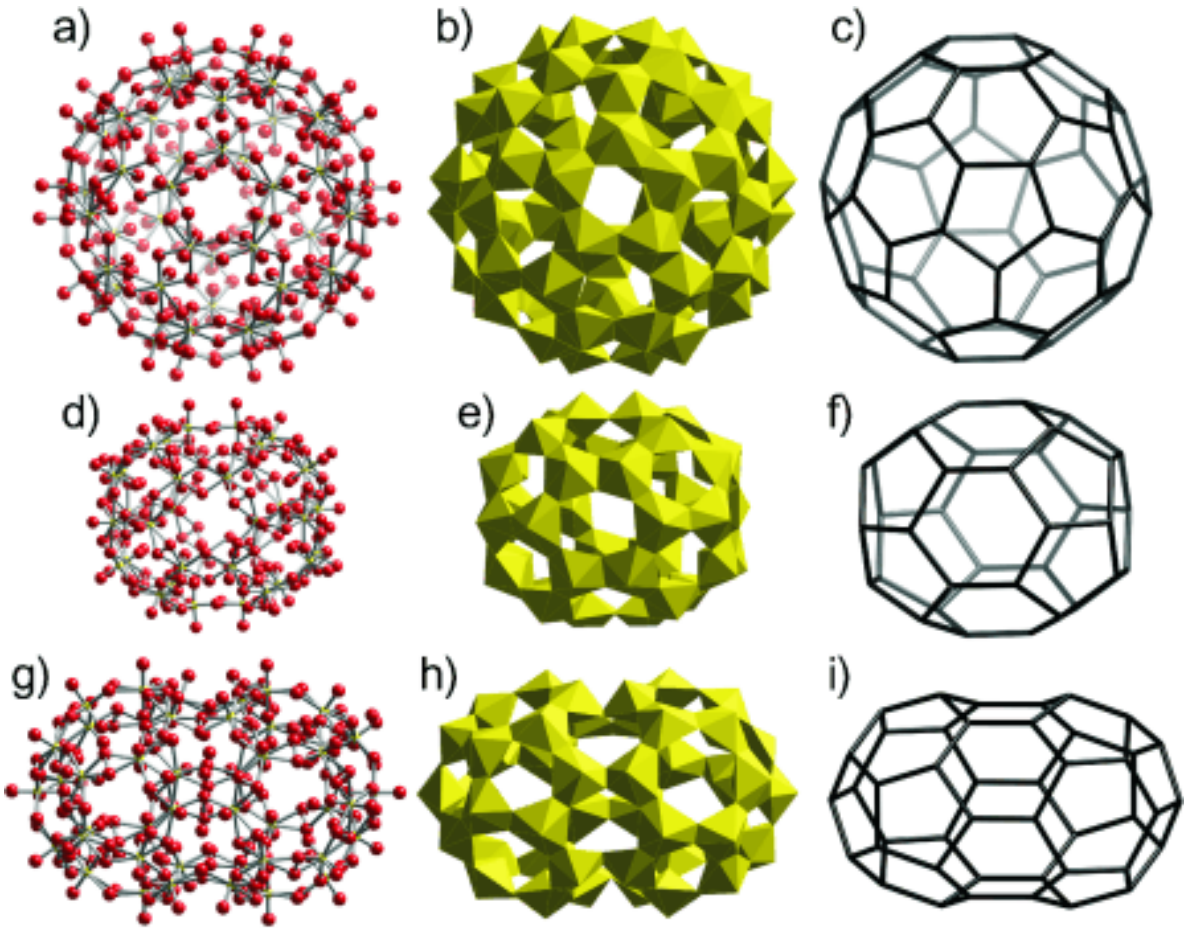


Geim, Nature Mater., **6**, 183 (2007)

- Future applications of graphene

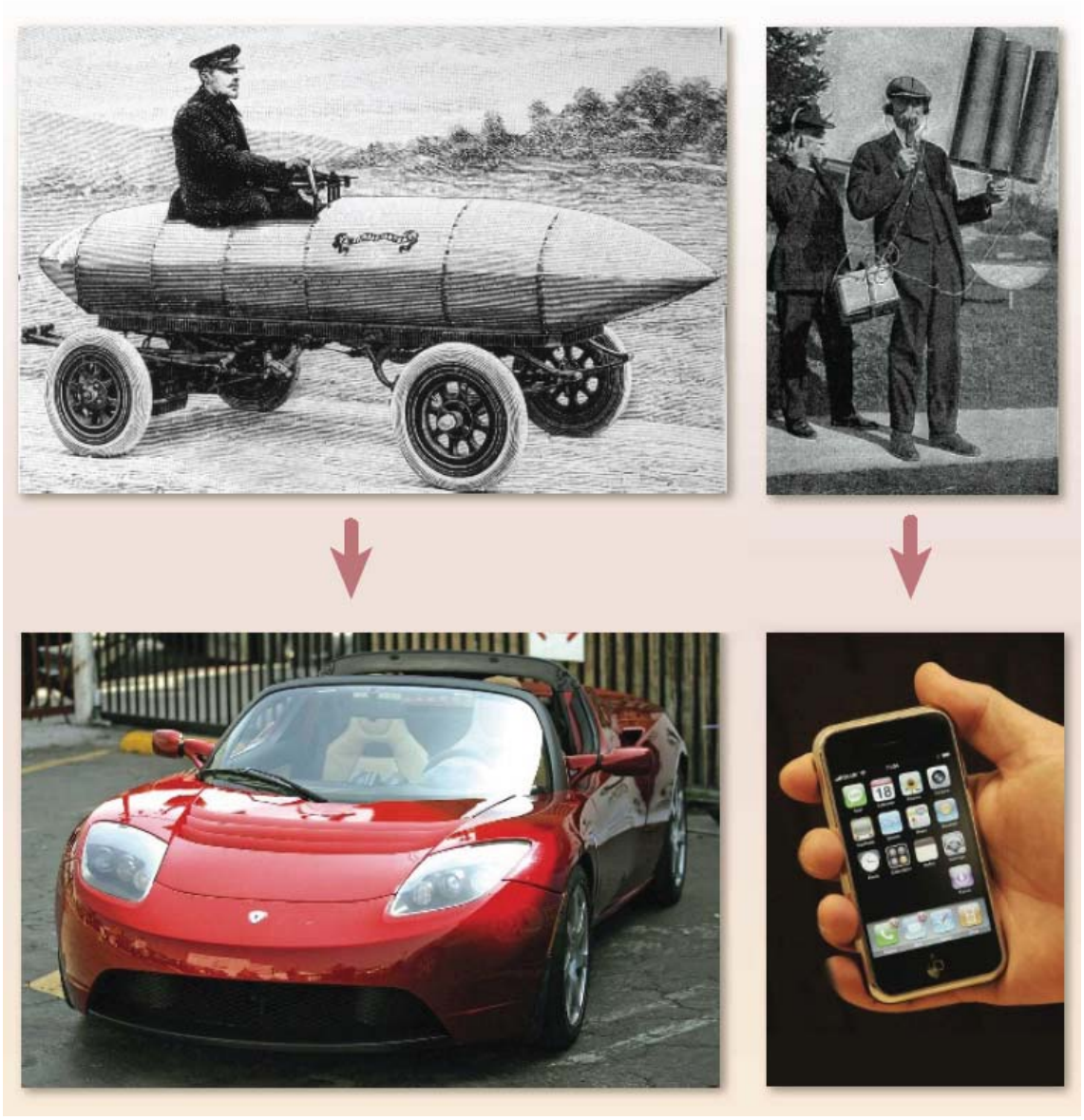


Uranyl Peroxide Fullerene



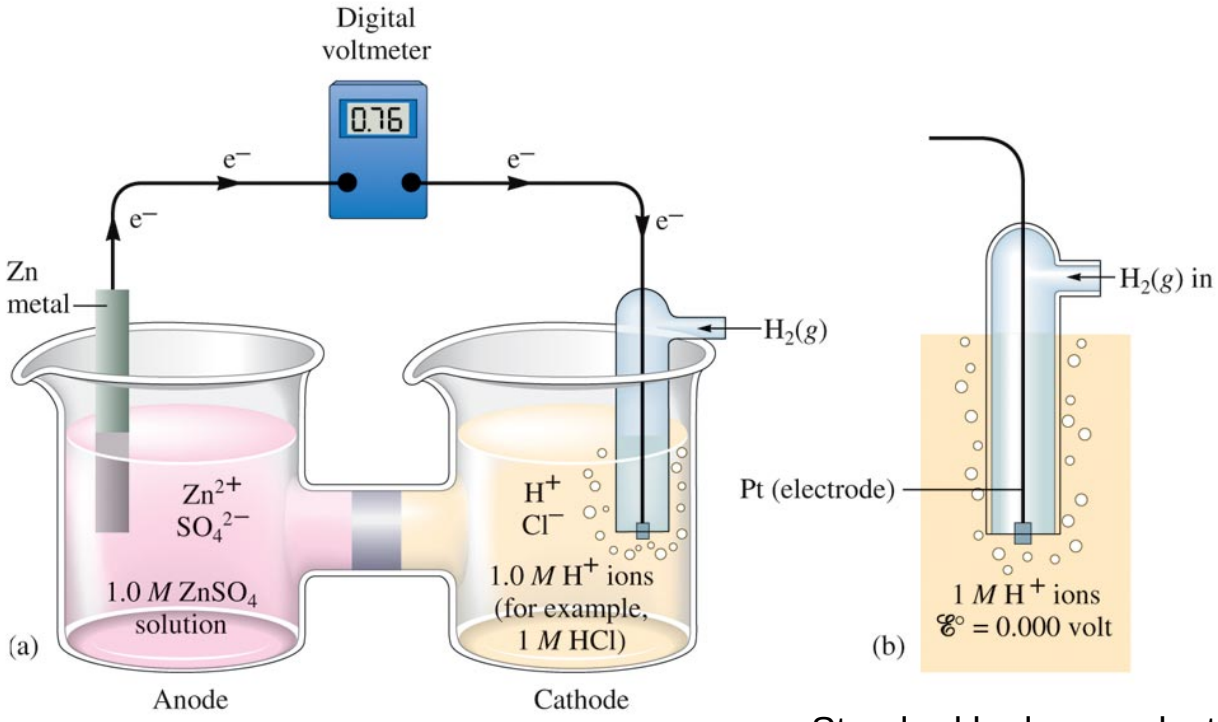
Sigmon et al., Angew. Chem. Int. Ed. **48**, 2737 (2009)

Building Better Batteries



Standard reduction potential

- Standard reduction potential



Standard hydrogen electrode (SHE)

Anode: $Zn \rightarrow Zn^{2+} + 2e^-$

Cathode : $2H^+ + 2e^- \rightarrow H_2$

Standard Reduction Potentials

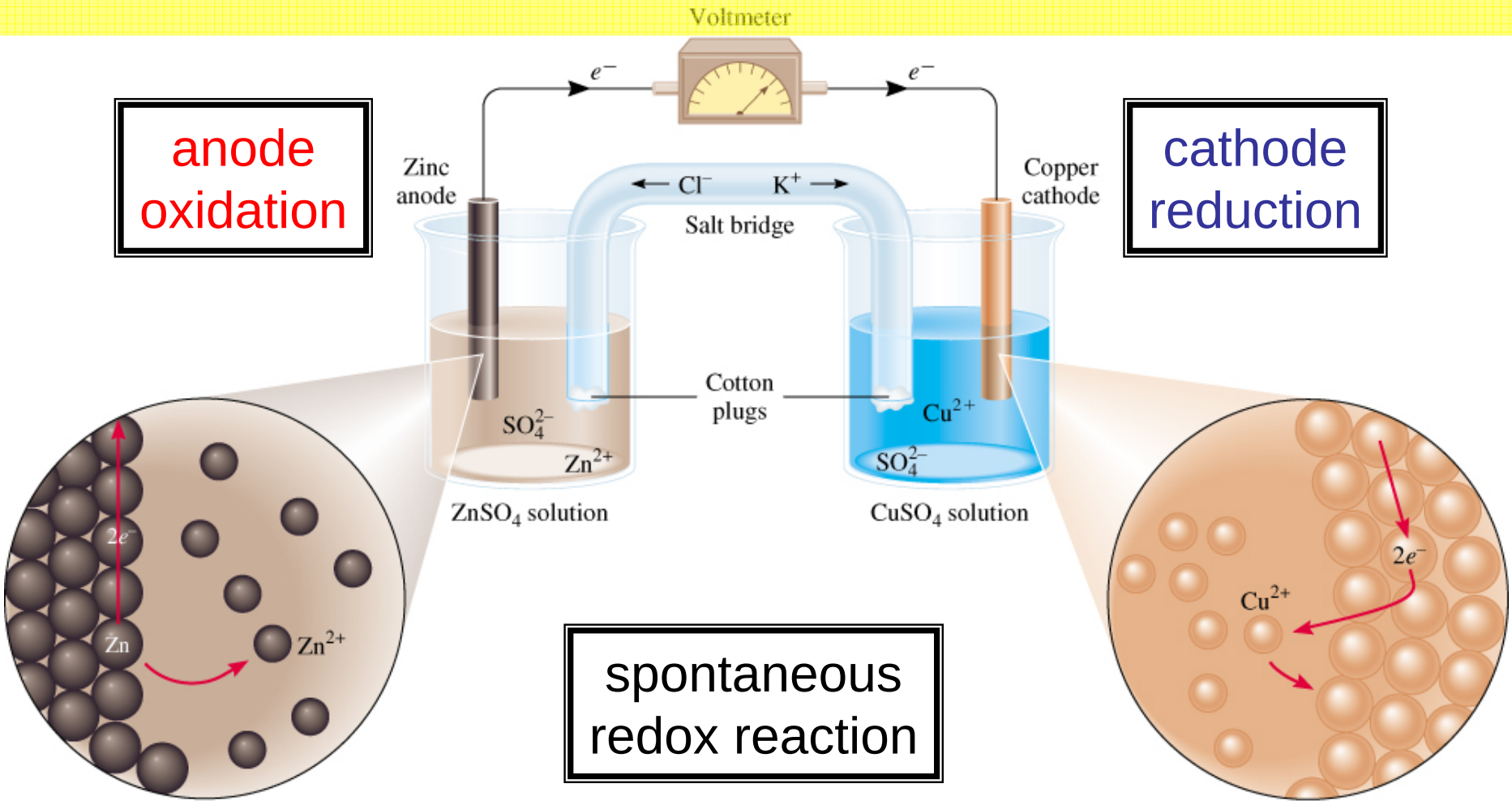
Increasing strength as oxidizing agent

TABLE 17.1 Standard Reduction Potentials at 25°C (298 K) for Many Common Half-Reactions

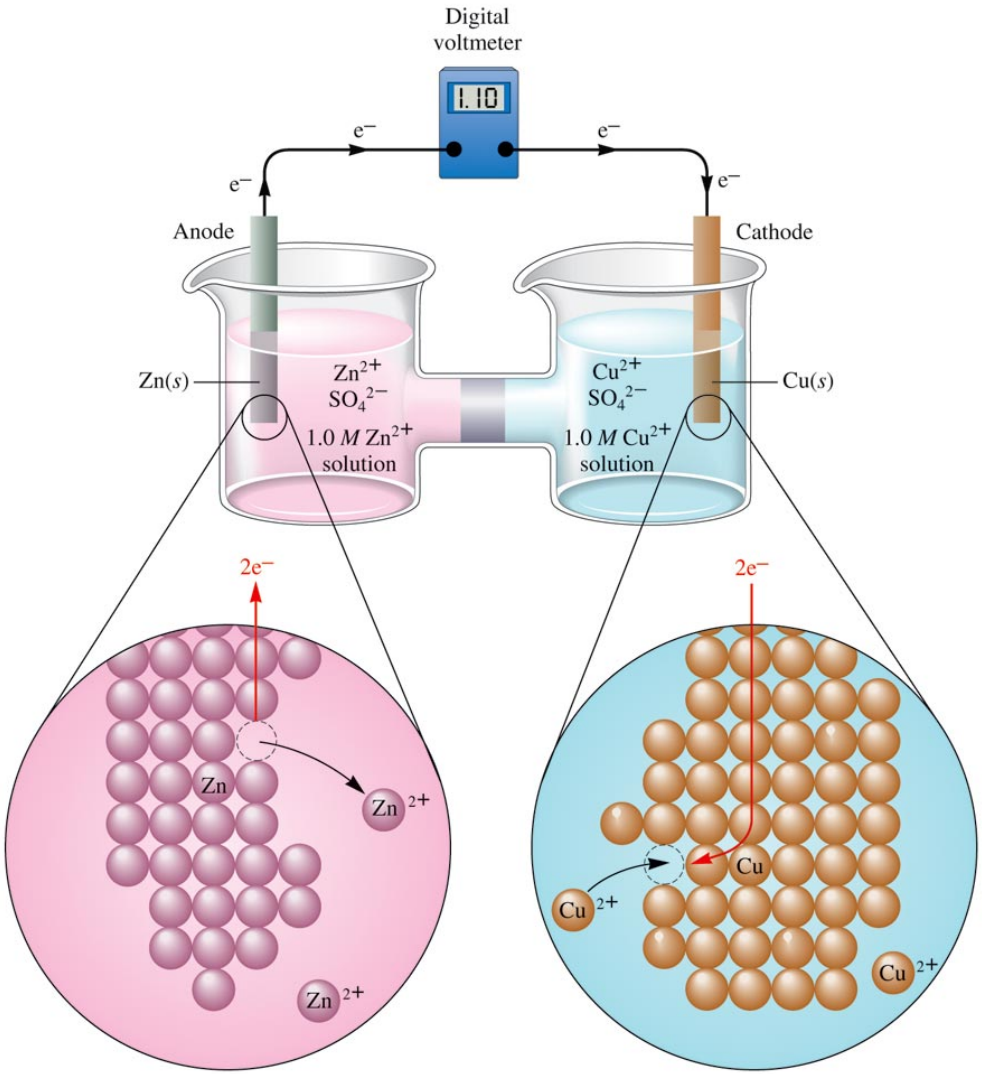
Half-Reaction	e° (V)	Half-Reaction	e° (V)
$F_2 + 2e^- \rightarrow 2F^-$	2.87	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	0.40
$Ag^{2+} + e^- \rightarrow Ag^+$	1.99	$Cu^{2+} + 2e^- \rightarrow Cu$	0.34
$Co^{3-} + e^- \rightarrow Co^{2-}$	1.82	$Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$	0.27
$H_2O_2 + 2H^- + 2e^- \rightarrow 2H_2O$	1.78	$AgCl + e^- \rightarrow Ag + Cl^-$	0.22
$Ce^{4+} + e^- \rightarrow Ce^{3+}$	1.70	$SO_4^{2-} + 4H^+ + 2e^- \rightarrow H_2SO_3 + H_2O$	0.20
$PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$	1.69	$Cu^{2+} + e^- \rightarrow Cu^+$	0.16
$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$	1.68	$2H^+ + 2e^- \rightarrow H_2$	0.00
$2e^- + 2H^+ + IO_4^- \rightarrow IO_3^- + H_2O$	1.60	$Fe^{3+} + 3e^- \rightarrow Fe$	−0.036
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	1.51	$Pb^{2+} + 2e^- \rightarrow Pb$	−0.13
$Au^{3+} + 3e^- \rightarrow Au$	1.50	$Sn^{2+} + 2e^- \rightarrow Sn$	−0.14
$PbO_2 + 4H^+ + 2e^- \rightarrow Pb^{2+} + 2H_2O$	1.46	$Ni^{2+} + 2e^- \rightarrow Ni$	−0.23
$Cl_2 + 2e^- \rightarrow 2Cl^-$	1.36	$PbSO_4 + 2e^- \rightarrow Pb + SO_4^{2-}$	−0.35
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	1.33	$Cd^{2+} + 2e^- \rightarrow Cd$	−0.40
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	1.23	$Fe^{2+} + 2e^- \rightarrow Fe$	−0.44
$MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$	1.21	$Cr^{3+} + e^- \rightarrow Cr^{2+}$	−0.50
$IO_3^- + 6H^+ + 5e^- \rightarrow \frac{1}{2}I_2 + 3H_2O$	1.20	$Cr^{3+} + 3e^- \rightarrow Cr$	−0.73
$Br_2 + 2e^- \rightarrow 2Br^-$	1.09	$Zn^{2+} + 2e^- \rightarrow Zn$	−0.76
$VO_2^+ + 2H^+ + e^- \rightarrow VO^{2+} + H_2O$	1.00	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	−0.83
$AuCl_4^- + 3e^- \rightarrow Au + 4Cl^-$	0.99	$Mn^{2+} + 2e^- \rightarrow Mn$	−1.18
$NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O$	0.96	$Al^{3+} + 3e^- \rightarrow Al$	−1.66
$ClO_2 + e^- \rightarrow ClO_2^-$	0.954	$H_2 + 2e^- \rightarrow 2H^-$	−2.23
$2Hg^{2+} + 2e^- \rightarrow Hg_2^{2+}$	0.91	$Mg^{2+} + 2e^- \rightarrow Mg$	−2.37
$Ag^+ + e^- \rightarrow Ag$	0.80	$La^{3+} + 3e^- \rightarrow La$	−2.37
$Hg_2^{2+} + 2e^- \rightarrow 2Hg$	0.80	$Na^+ + e^- \rightarrow Na$	−2.71
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	0.77	$Ca^{2+} + 2e^- \rightarrow Ca$	−2.76
$O_2 + 2H^+ + 2e^- \rightarrow H_2O_2$	0.68	$Ba^{2+} + 2e^- \rightarrow Ba$	−2.90
$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	0.56	$K^+ + e^- \rightarrow K$	−2.92
$I_2 + 2e^- \rightarrow 2I^-$	0.54	$Li^+ + e^- \rightarrow Li$	−3.05
$Cu^+ + e^- \rightarrow Cu$	0.52		

Increasing strength as reducing agent

Galvanic Cells



Galvanic Cells



Anode: $Zn \rightarrow Zn^{2+} + 2e^-$

Cathode : $Cu^{2+} + 2e^- \rightarrow Cu$

$Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$

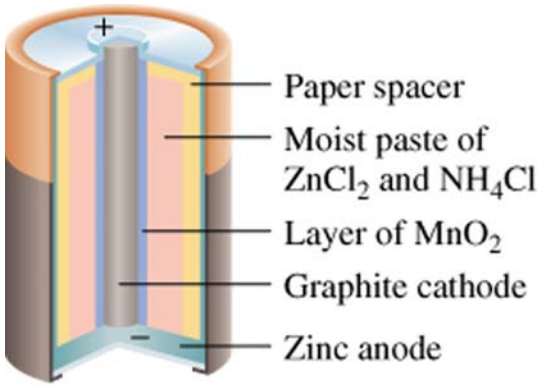
$$E^\circ = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.34\ \text{V} - (-0.76\ \text{V}) = 0.10\ \text{V}$$

Batteries

- Battery: a galvanic cell, or a series of combined galvanic cells, that can be used as a source of direct electric current at a constant voltage.

  - The dry cell battery
  - The mercury battery
  - The lead storage battery
  - The solid-state lithium battery
  - Fuel Cells

The Dry Cell Battery



Leclanché cell

Anode: $Zn\ (s) \longrightarrow Zn^{2+}\ (aq) + 2e^-$

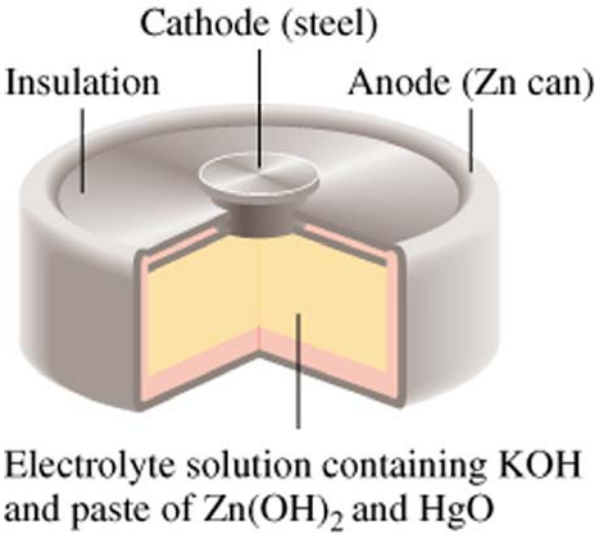
Cathode: $2NH_4^+\ (aq) + 2MnO_2\ (s) + 2e^- \longrightarrow Mn_2O_3\ (s) + 2NH_3\ (aq) + H_2O\ (l)$

Alkaline dry cell: the solid NH_4Cl is replaced with KOH or NaOH.

Anode: $Zn\ (s) + OH^-(aq) \longrightarrow ZnO\ (s) + H_2O\ (l) + 2e^-$

Cathode: $2MnO_2\ (s) + H_2O\ (l) + 2e^- \longrightarrow Mn_2O_3\ (s) + 2OH^-\ (aq)$

The Mercury Battery

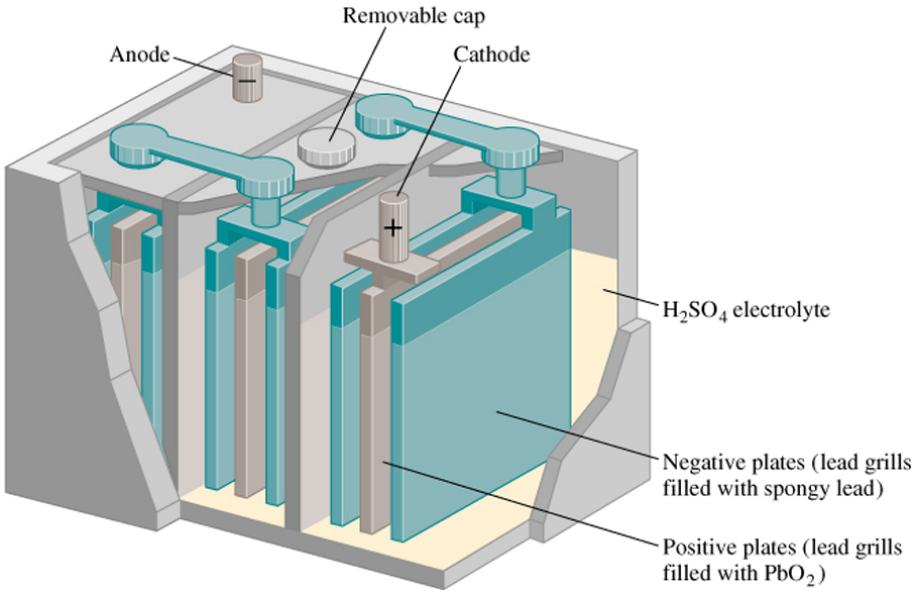


Anode: $Zn(Hg) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(l) + 2e^-$

Cathode: $HgO(s) + H_2O(l) + 2e^- \longrightarrow Hg(l) + 2OH^-(aq)$

$$Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(l)$$

The Lead Storage Battery

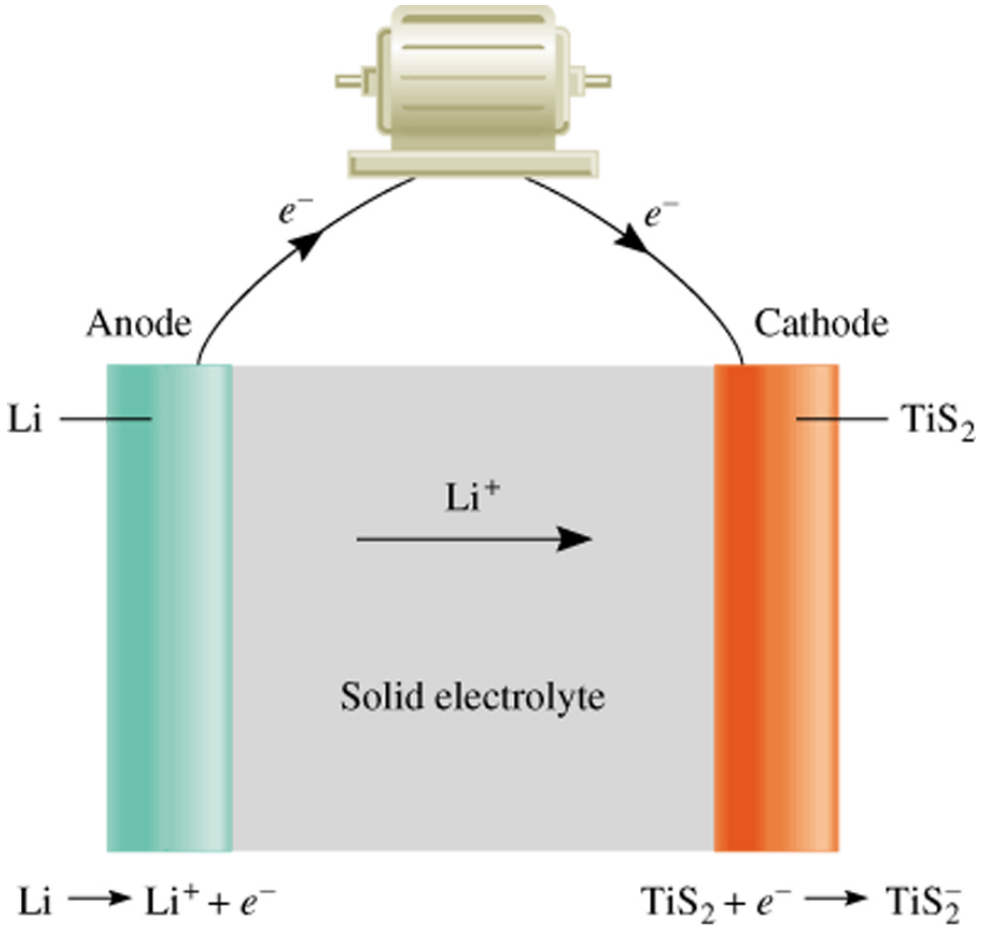


Anode: $Pb\ (s) + SO_4^{2-}\ (aq) \longrightarrow PbSO_4\ (s) + 2e^-$

Cathode: $PbO_2\ (s) + 4H^+\ (aq) + SO_4^{2-}\ (aq) + 2e^- \longrightarrow PbSO_4\ (s) + 2H_2O\ (l)$

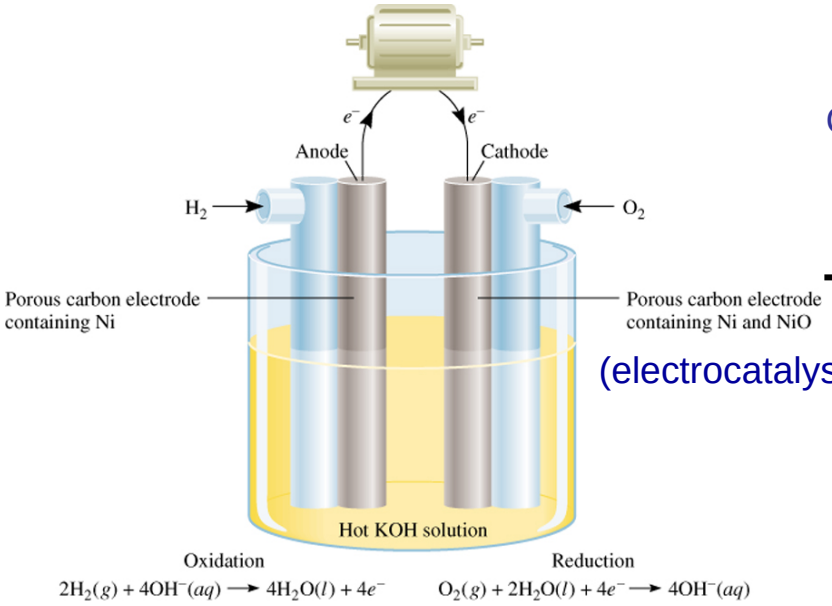
$$Pb\ (s) + PbO_2\ (s) + 4H^+\ (aq) + 2SO_4^{2-}\ (aq) \longrightarrow 2PbSO_4\ (s) + 2H_2O\ (l)$$

The Solid-State Lithium Battery



Fuel Cells

- Fuel cells: a electrochemical (=galvanic) cell that requires a continuous supply of reactants to keep functioning
- Conversion of fossil fuel into electrical energy is a highly inefficient process ~ only 40 %



Anode: $2H_2\,(g) + 4OH^-\,(aq) \longrightarrow 4H_2O\,(l) + 4e^-$

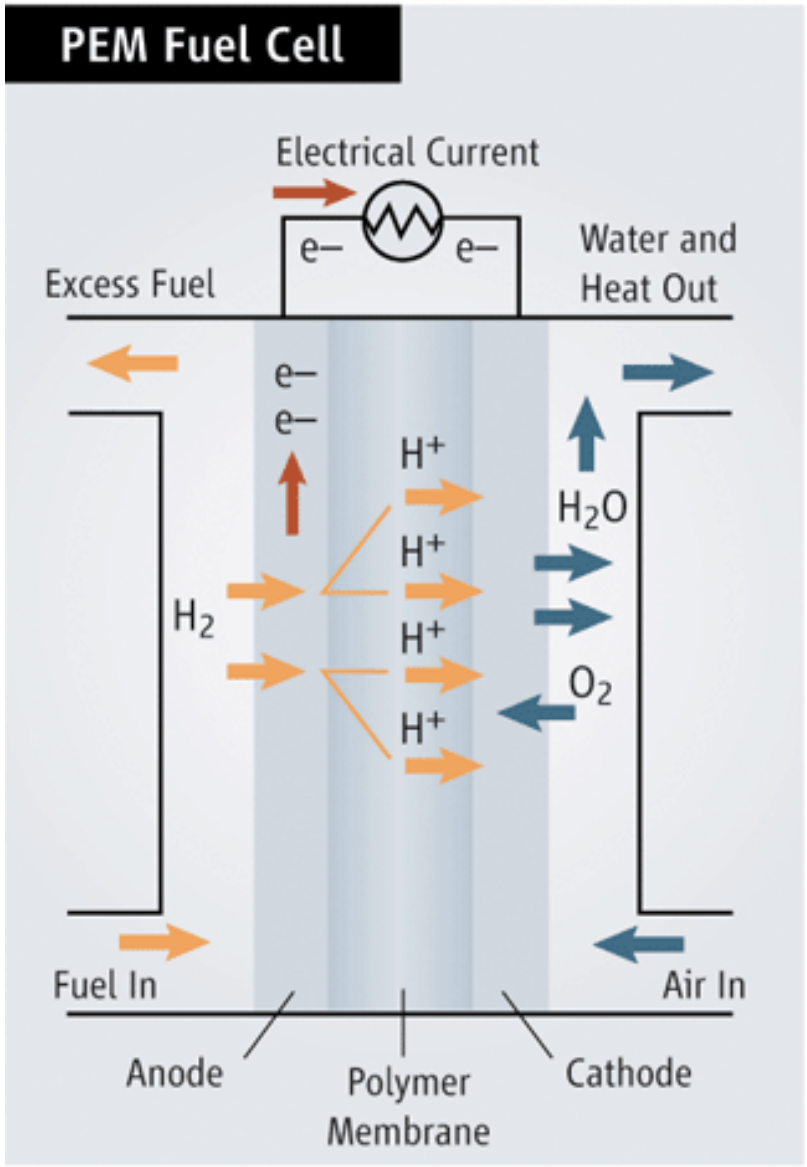
Cathode: $O_2\,(g) + 2H_2O\,(l) + 4e^- \longrightarrow 4OH^-\,(aq)$

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O\,(l)$$

$$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.40\ \text{V} - (-1.23\ \text{V}) = 1.23\ \text{V}$$

A major problem of Fuel cells lies in the lack of cheap electrocatalysts able to function efficiently for long periods of time with contamination.

Proton Exchange Membrane (PEM) in Fuel Cells



At one electrode, hydrogen molecules are stripped of their electrons, which are then sent through an external circuit to do work. The leftover protons are channeled through the polymer membrane to another electrode, where they meet up with oxygen and the circulating electrons to produce water.

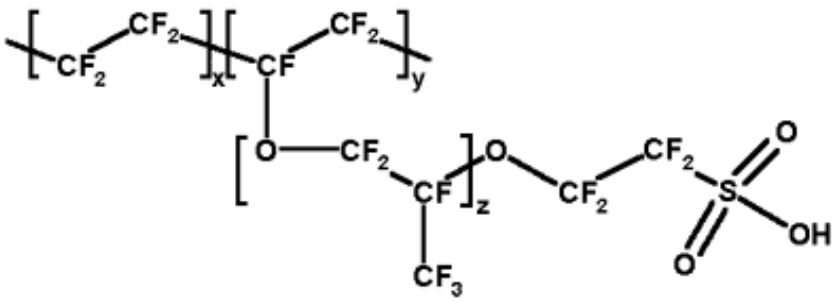
Anode: $2H_2\,(g) \longrightarrow 4H^+\,(aq) + 4e^-$

Cathode: $O_2\,(g) + 4H^+\,(aq) + 4e^- \longrightarrow 2H_2O\,(l)$

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O\,(l)$$

$$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.0\ \text{V} - (-0.83\ \text{V}) = 1.23\ \text{V}$$

Nafion as a Polymer Membrane



- Nafion
  1. sulfonated tetrafluoroethylene based fluoropolymer copolymer
  2. Made by DuPont
  3. Ionomer → conductivity
  4. Excellent thermal and mechanical stability.
  5. Protons on the SO_3H (sulfonic acid) groups "hop" from one acid site to another.
  6. Pores allow movement of cations but the membranes do not conduct anions or electrons.

Ionomer: a polymer that comprises repeat units of both eleectrically neutral repeating units and a fraction of ionized units (usually no more than 15 percent).

Secondary Electrochemical Cells: Rechargeable Batteries

- Ni-metal hydride batteries (Ni-MH) (M =





- Lithium ion batteries





Ni-MH batteries

- M = AB_5 where A is a rare earth mixture of lanthanum, cerium, neodymium, praseodymium and B is nickel, cobalt, manganese, and/or aluminium.
- Discharging

Anode: $MH + OH^- \longrightarrow M + H_2O + e^-$

Cathode: $NiO(OH) + H_2O\ (aq) + e^- \longrightarrow Ni(OH)_2 + OH^-$

$NiO(OH) + MH \longrightarrow Ni(OH)_2 + M$

- Charging

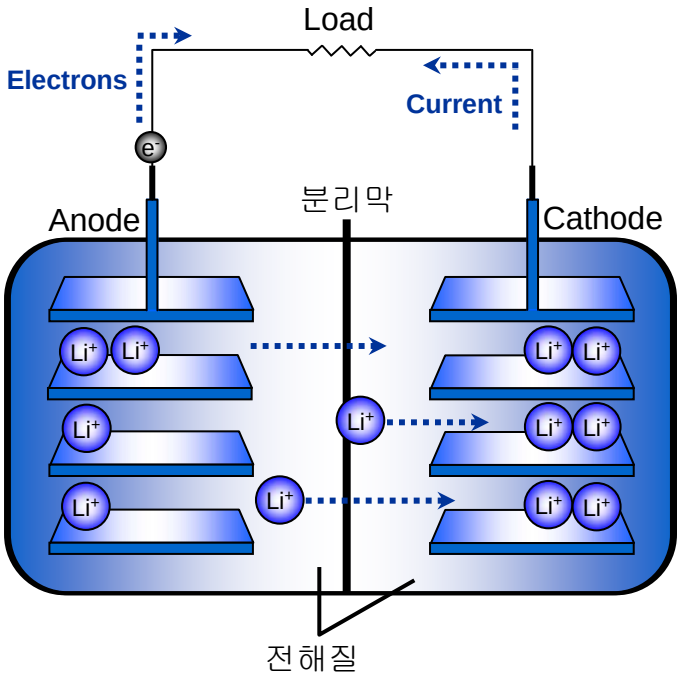
Anode: $M + H_2O + e^- \longrightarrow MH + OH^-$

Cathode: $Ni(OH)_2 + OH^- \longrightarrow NiO(OH) + H_2O\ (aq) + e^-$

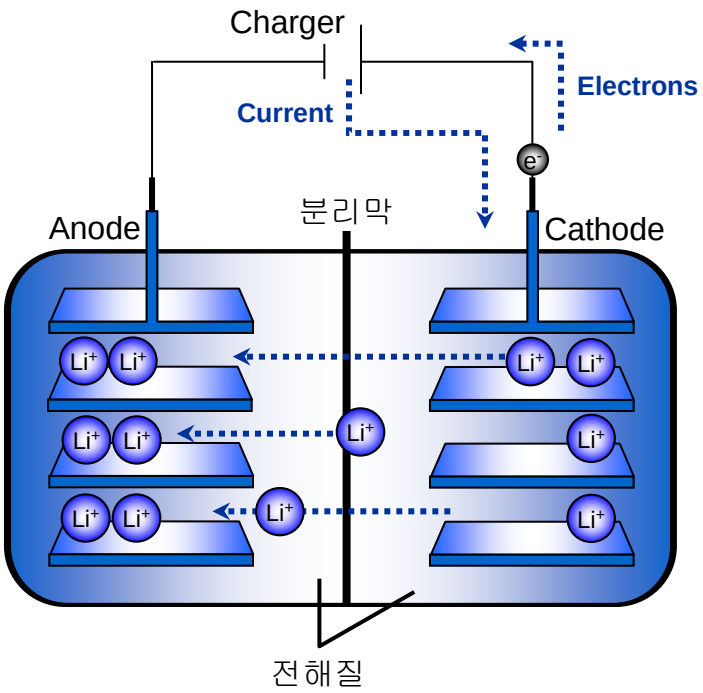
$Ni(OH)_2 + M \longrightarrow NiO(OH) + MH$

Mechanism of Lithium Ion Batteries

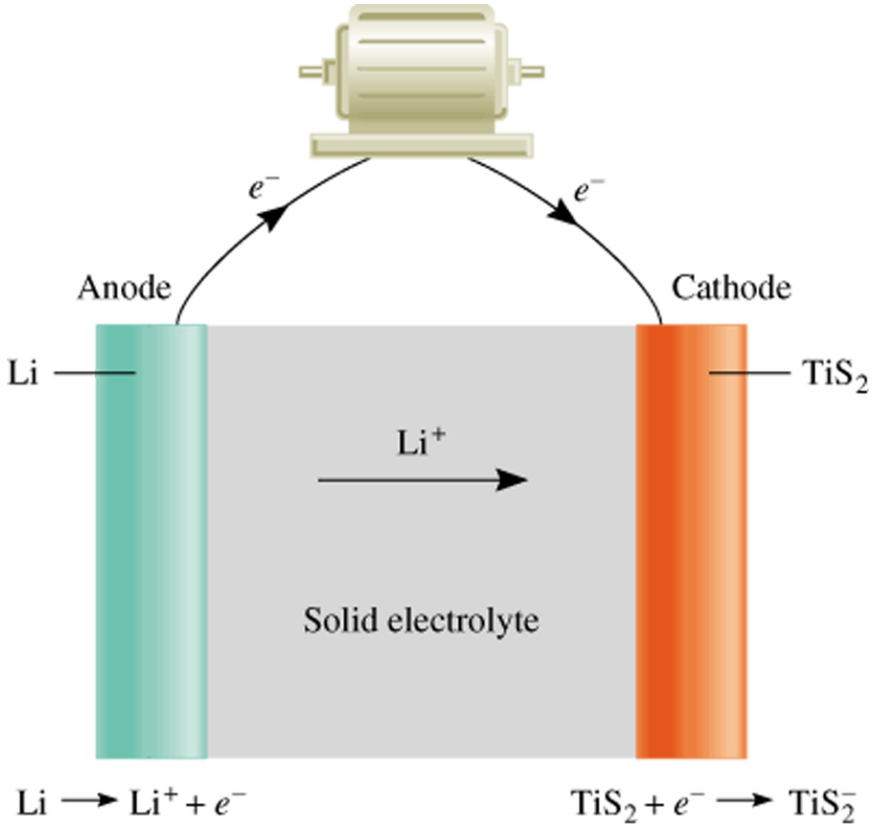
방전 반응 **(Discharging)**



충전 반응 **(Charging)**



The Original Lithium Ion Battery



Modern Lithium Ion Batteries

- Discharging

Anode: $Li_xC_6 \rightarrow xLi^+ + 6C + xe^-$

Cathode: $Li_{1-x}CoO_2 + xLi^+ + xe \rightarrow LiCoO_2$

$$Li_{1-x}CoO_2 + Li_xC_6 \rightarrow LiCoO_2 + 6C$$

- Charging

Anode: $xLi^+ + 6C + xe^- \rightarrow Li_xC_6$

Cathode: $LiCoO2 \rightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$

$$LiCoO_2 + 6C \rightarrow Li_{1-x}CoO_2 + Li_xC_6$$

리튬-이온 2차 전지의 장·단점

장 점

- 에너지 밀도가 높다.
- 작동 전압이 높다.
- 무공해
- 우수한 수명
- 메모리 효과가 없다.

리튬-이온 2차 전지의 장·단점

단 점

- 외부 충격에 약하다.
- 80℃ 이상의 고열이 가해지면 단락이 발생할 수 있다.
- 휘발성이 있는 유기성 전해액이 사용된다.
- 코발트가 지구상에 소량으로 존재

리튬-이온 2차 전지의 응용

- 휴대폰, **MP3,** 노트북 등 휴대용 **IT**기기





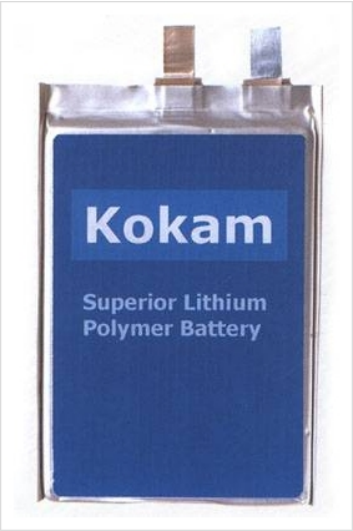


- **HEV (Hybrid Electric Vehicle)**



전지의 최근 이슈

- 리튬폴리머 전지의 개발
  - 전기 전도도가 우수한 폴리머 전해질 사용
  - 리튬-이온 전지보다 안전
  - 다양한 모양의 전지를 개발

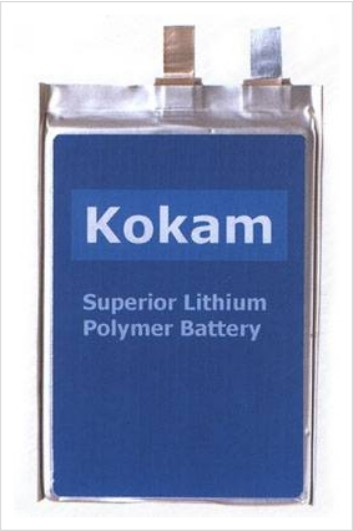


- **HEV (Hybrid Electric Vehicle)**의 개발



전지의 최근 이슈

- 리튬폴리머 전지의 개발
  - 전기 전도도가 우수한 폴리머 전해질 사용
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- **HEV (Hybrid Electric Vehicle)**의 개발



리튬-폴리머 전지의 최근 이슈

스웰링 현상(임신 전지)



전지의 최근 이슈

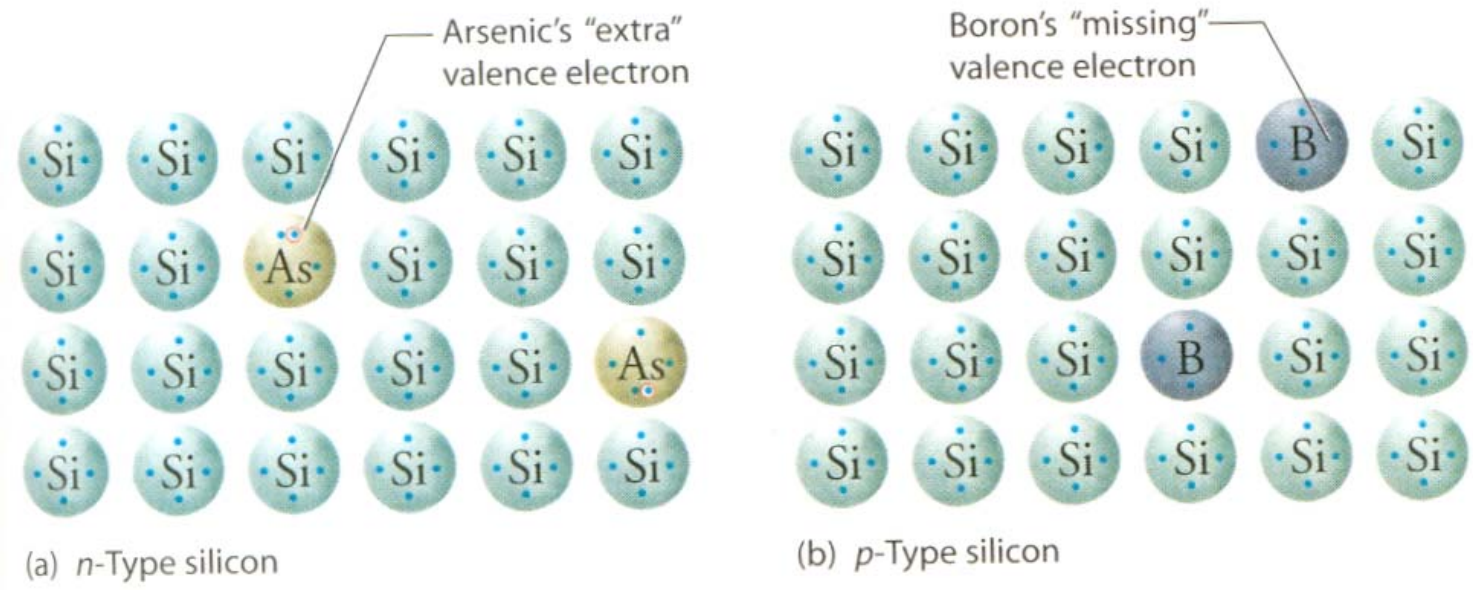
배터리 폭발





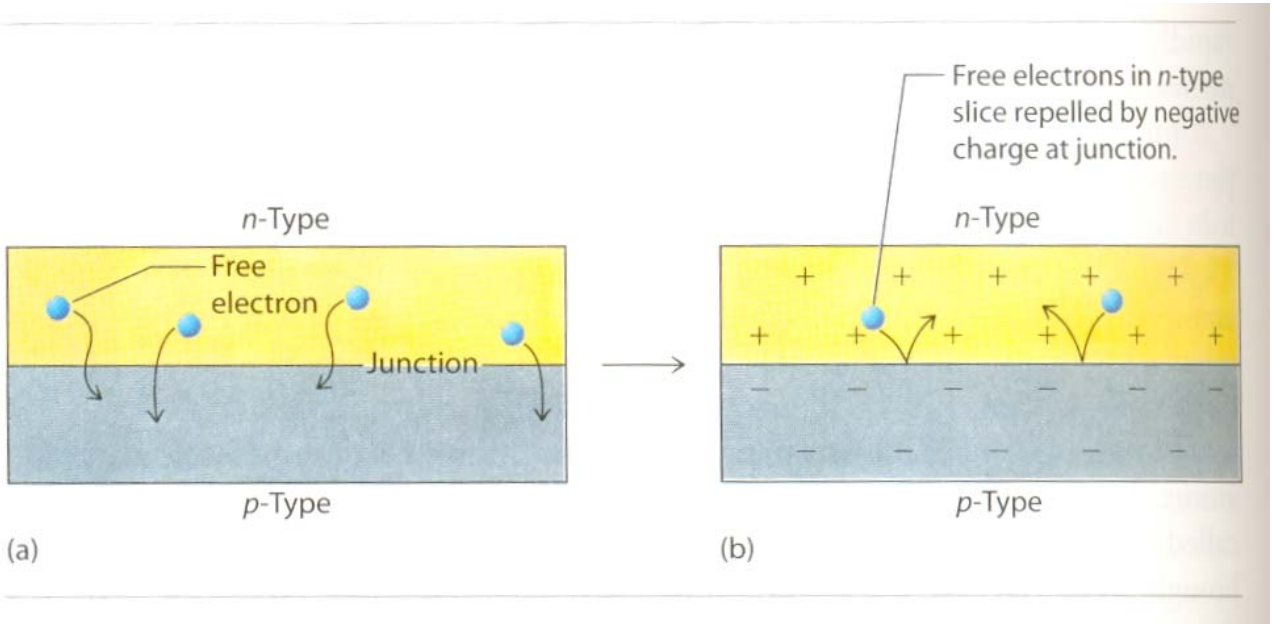
Solar Cells: A Photovoltaic Cell

- Photovoltaic cells: a device that converts the energy of sunlight into electricity.
- Semiconductor-based solar cells:



Silicon-Based Solar Cells

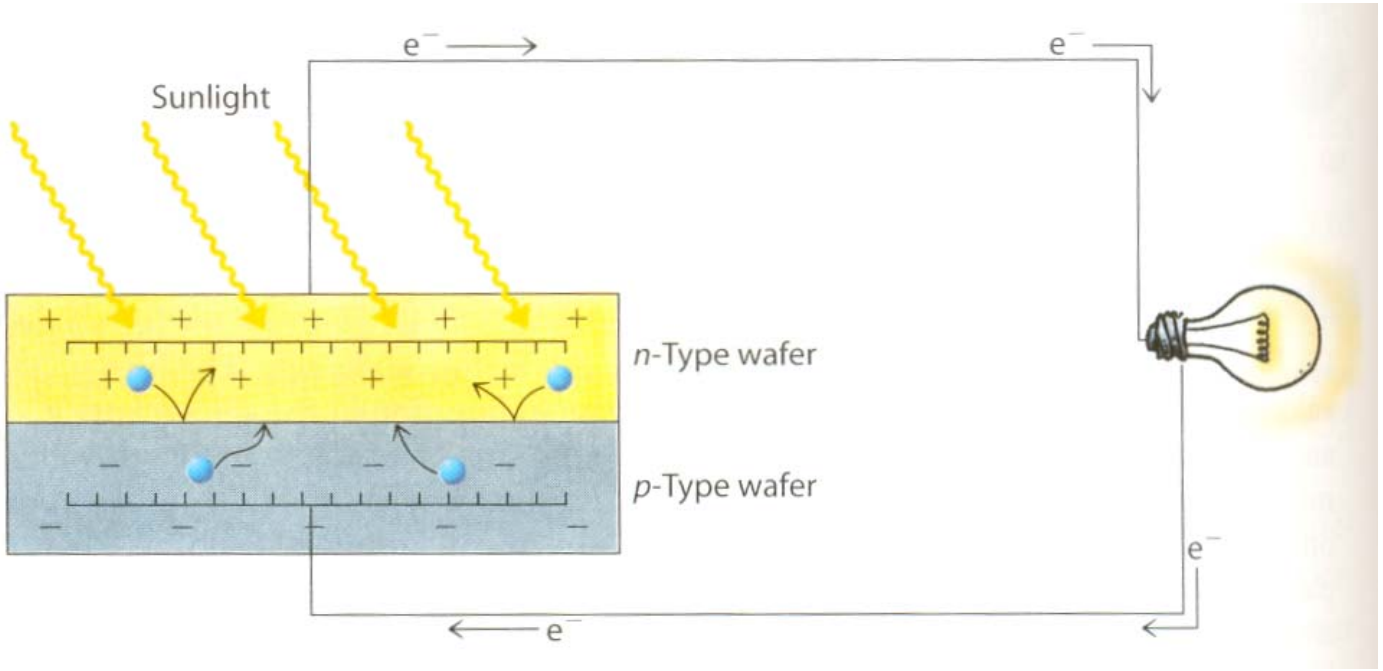
- p-n junction



- (a) Initial, free electrons migrate from the n-type silicon to the p-type silicon.
- (b) Quickly, however, charge buildup at the junction prevents the continued flow of electrons.

Silicon-Based Solar Cells

- Free electrons in the n-type wafer are repelled by the positive charge of the p-type wafer. Free electrons in the p-type wafer, however, are attracted to the positive charge of the n-type wafer. The energy of sunlight gets these electrons moving to the point they are pushed into an external circuit in a unidirectional fashion.



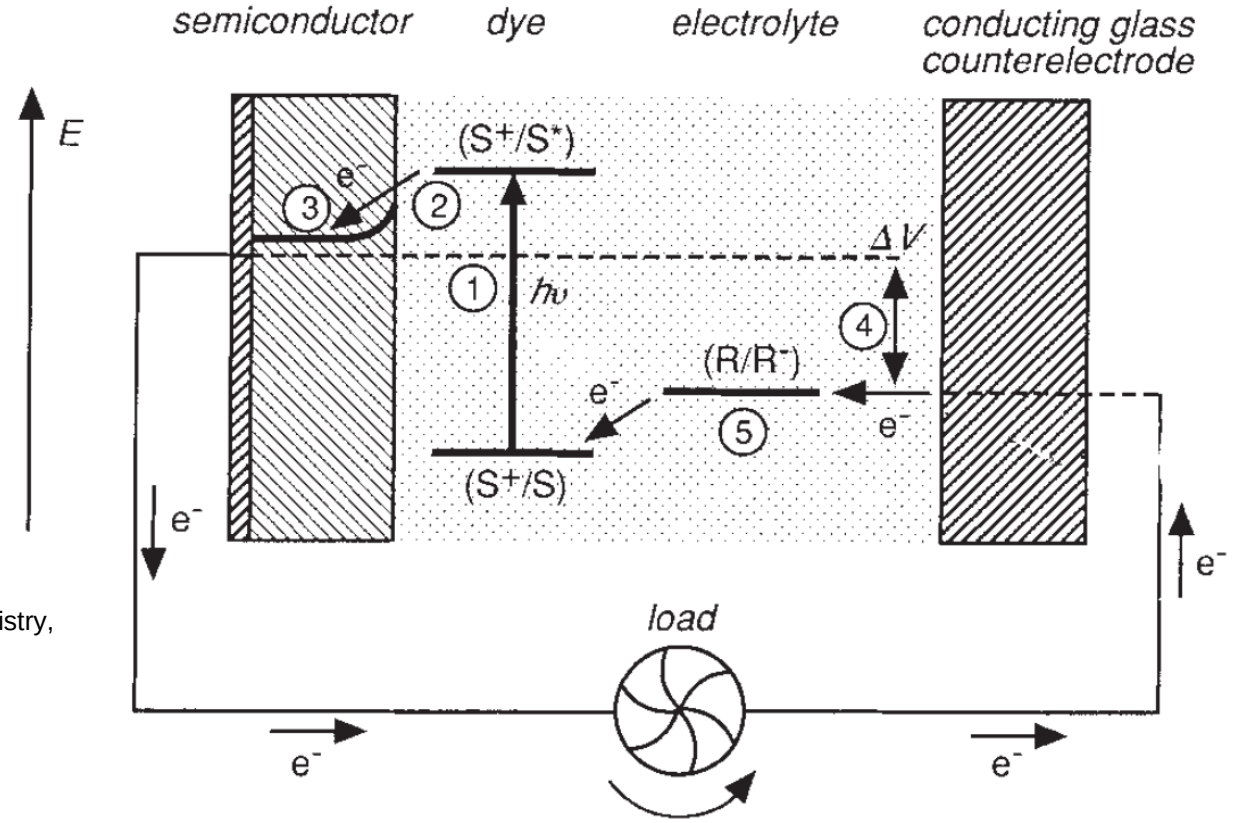
- The energy conversion efficiency of silicon-based solar cells are about 15 %.

Dye-Sensitized Solar Cells (DSSC)

- Dye-Sensitized Solar Cells are also called Grätzel cells



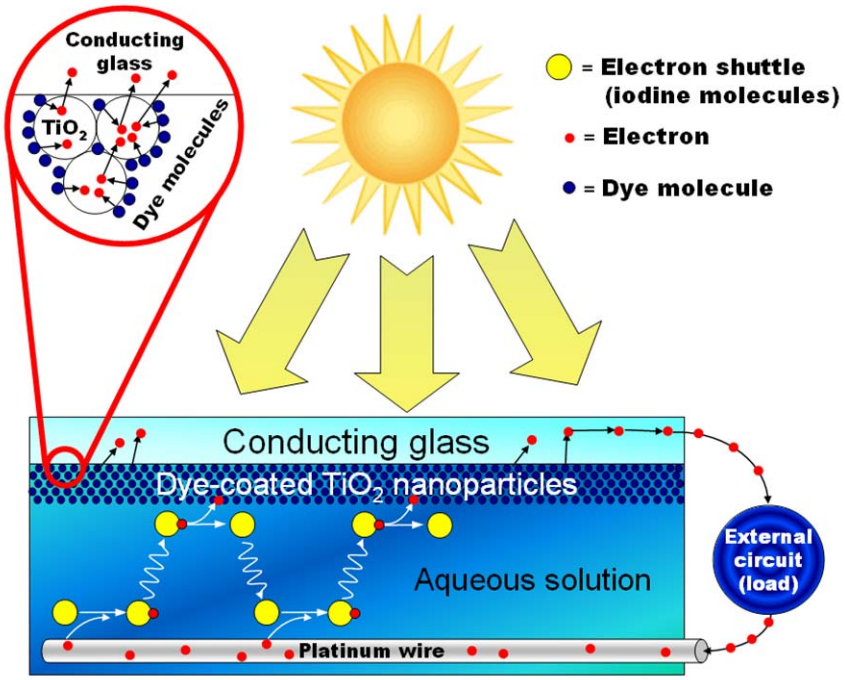
Professor at the Institute of Physical Chemistry,
Swiss Federal Institute of Technology,
Lausanne, Switzerland.

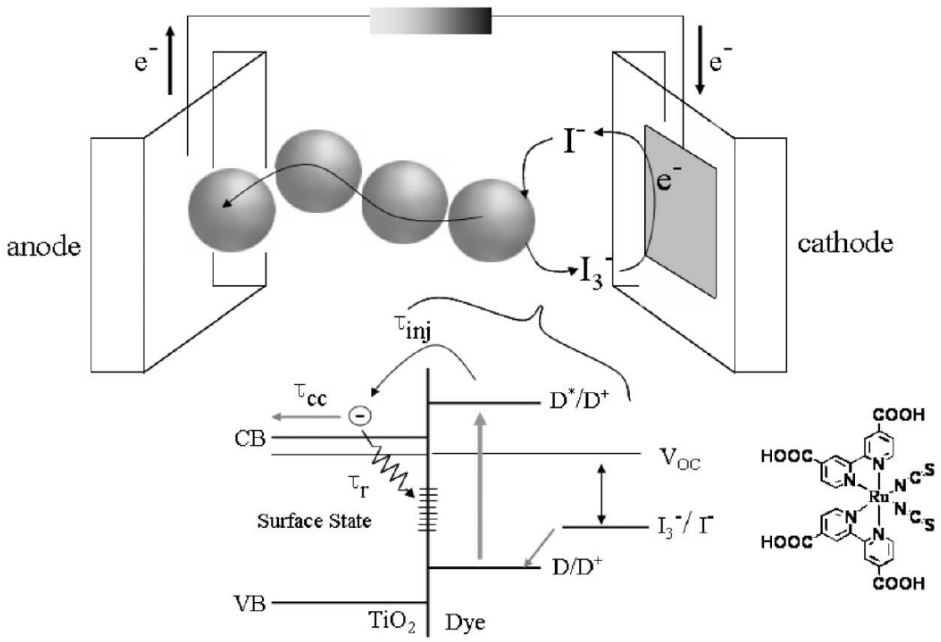


Gratzel et al., Nature **353**, 737 (1991)

Dye-Sensitized Solar Cells (DSSC)

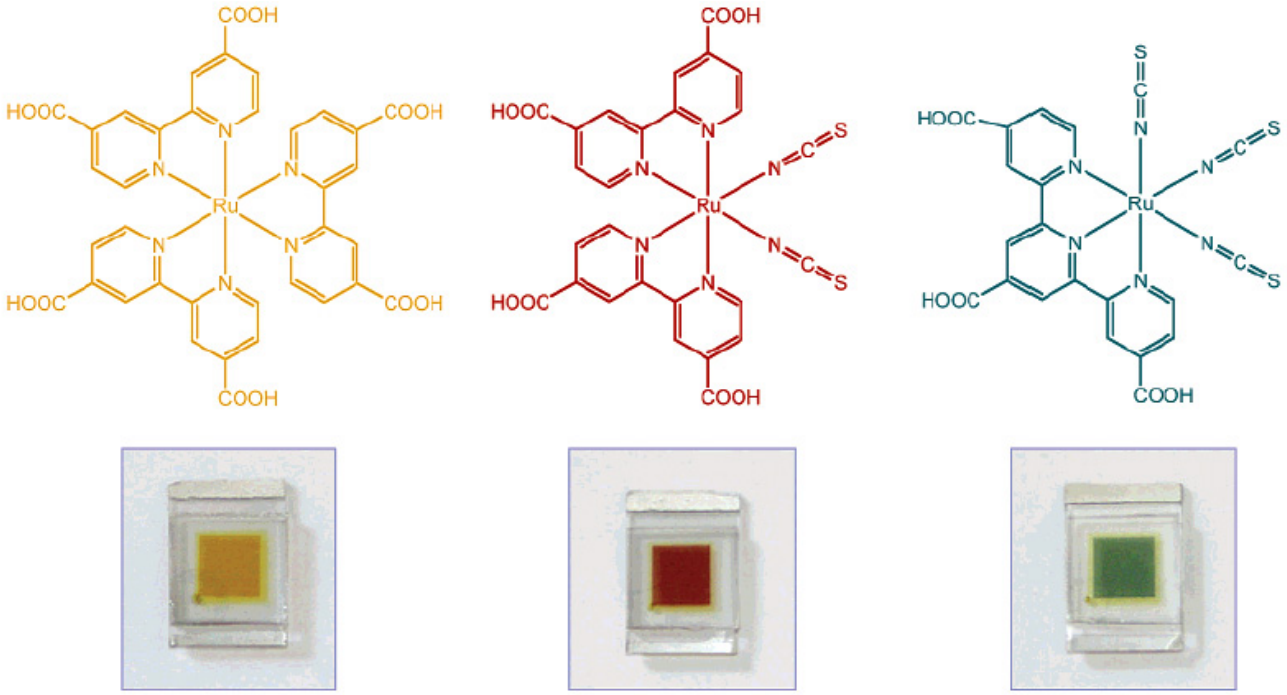
- A schematic of the dye-sensitized solar cell





Dye-Sensitized Solar Cells (DSSC)

- Ruthenium-based dyes



Ruthenium sensitizers RuL_3 (yellow) cis-$RuL_2(NCS)_2$ (red) and $RuL'(NCS)_3$ (green)
where L = 2,2'-bipyridyl-4,4'-dicarboxylic acid and L' = 2,2',2''-terpyridyl-4,4',4''-tricarboxylic acid.

Dye-Sensitized Solar Cells (DSSC)

- Organic molecule-based dyes

